20 h is inversely proportional to p
p is directly proportional to $\sqrt{t}$

Given that $h = 10$ and $t = 144$ when $p = 6$
find a formula for h in terms of t

h is inverse to p, so h = A/p.

1 mark
Set up inverse proportion.

10 = A/6, so A = 60.

1 mark
Use given h and p.

p is direct to √t, so p = B√t.

6 = B√144 = 12B, so B = 1/2.

1 mark
Use p and t to find B.

Then h = 60/((1/2)√t) = 120/√t.

1 mark
Formula h = 120/√t.

$h = \dfrac{120}{\sqrt{t}}$

(Total for Question 20 is 4 marks)

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